Windows Registry Forensics

Tasks

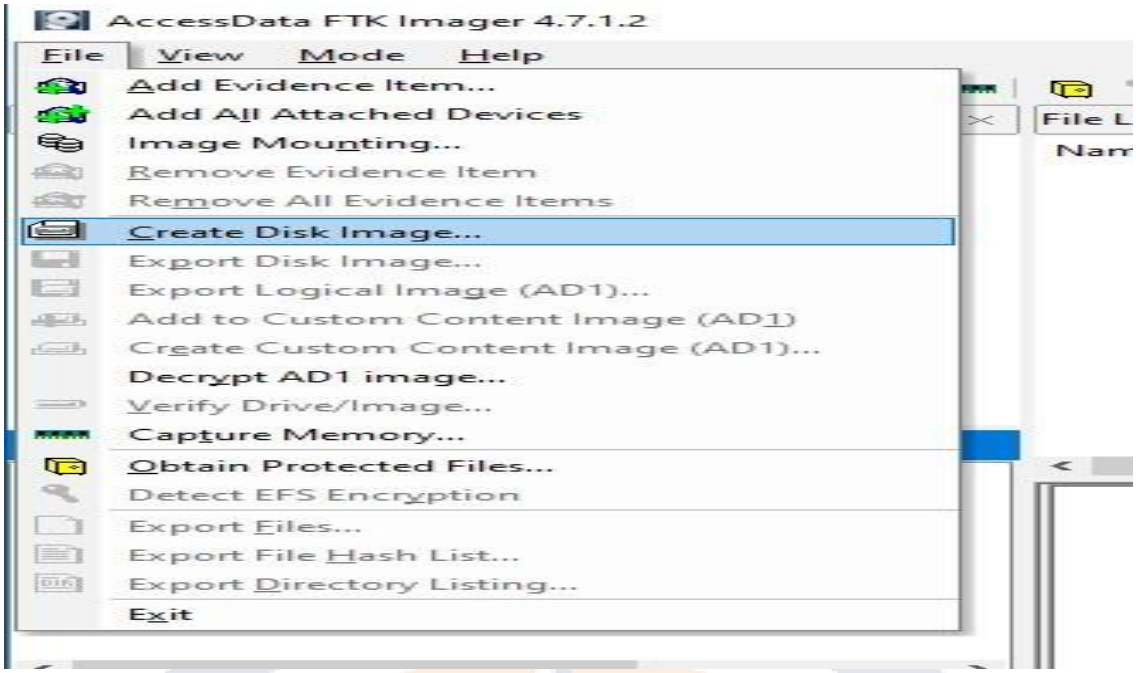
- Acquiring Windows Registry
- Analysing Extracted Registry Hives Using RegRipper
- Analysing extracted registry hives using registry explorer tool
- Hands-On Project
- Analyse the given image and answer the given questions

Acquiring Windows Registry

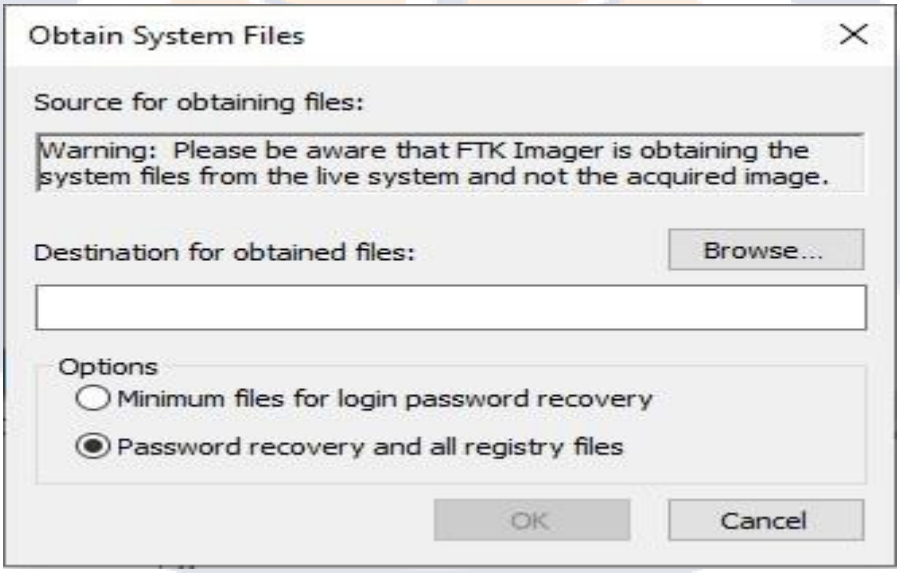
Computer forensics tools will acquire Windows registry files as a part of acquiring the target machine's system drive or when performing a complete hard drive acquisition. You can also extract only the registry files from a live system and store it separately for later analysis (this is usually referred as "Registry Image"). The following shows how we can do this using AccessData FTK Imager.

To acquire target Windows machine registry using FTK Imager, follow these steps:

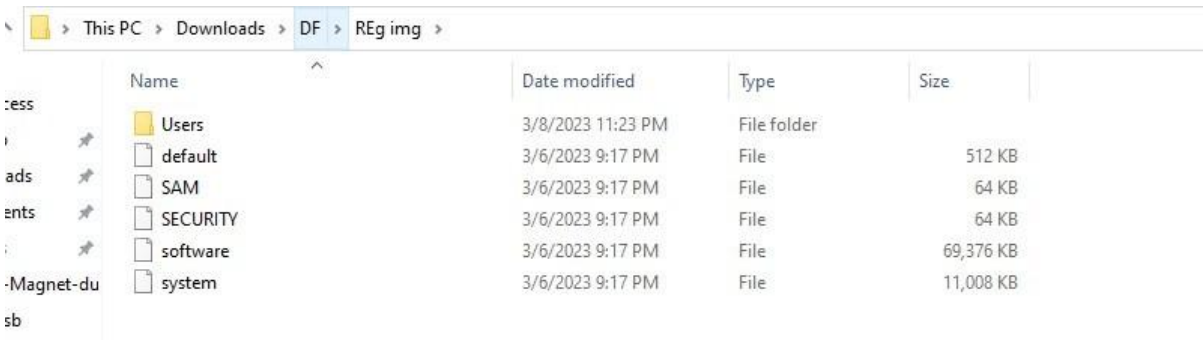
1. Generally we would download AccessData FTK Imager and transfer it into your USB thumb drive, in this case you can just run FTK Imager and obtain a registry image of your Windows machine.
2. Open FTK Image, and go to the File menu ➤ Obtain Protected Files…



3. A new dialog appears; select where you want to store obtained files, and check the option “Password recovery and all registry files”. Finally, click the “OK” button.



4. A progress window will appear showing registry files’ export progress; upon finishing, the window will disappear without announcing any success message. Go to the directory where you have saved your registry files to see the resultant files; you should see the five files and one folder.



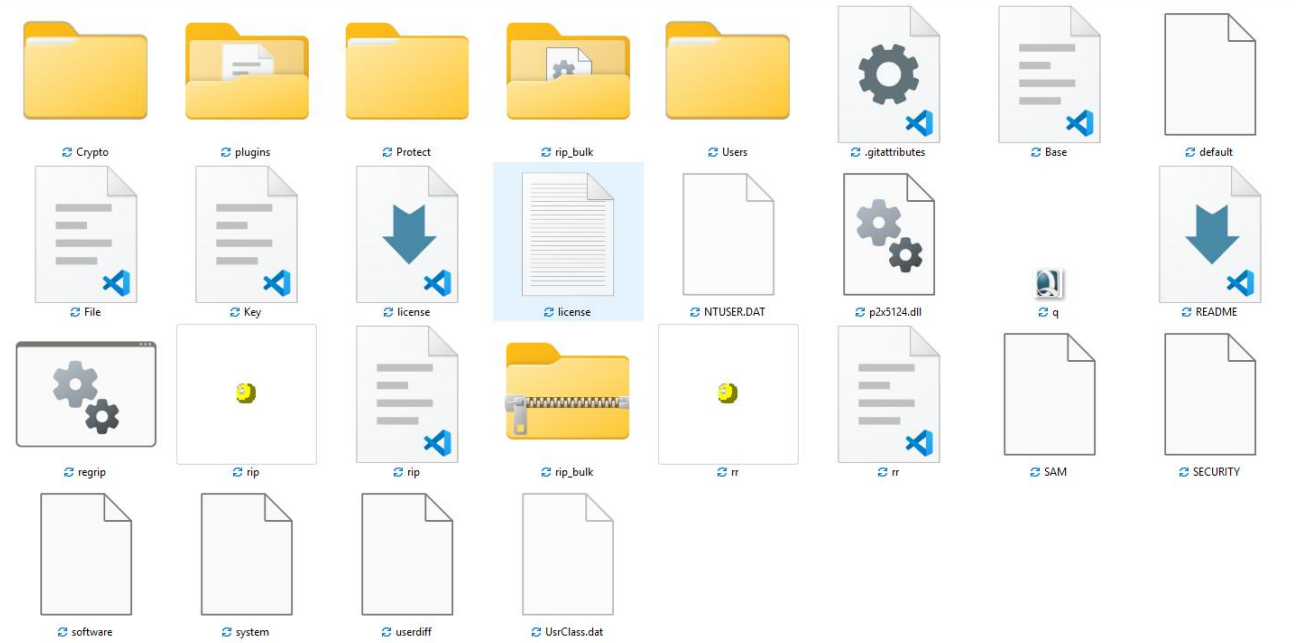
Analysing Extracted Registry Hives Using RegRipper

RegRipper is a forensic tool that automatically parses registry hives and extracts important artifacts using predefined plugins.

Steps to Analyse Registry Using RegRipper

Step 1: Setup RegRipper

1. Download and extract RegRipper.
2. Copy the extracted registry hive files (SYSTEM, SOFTWARE, NTUSER.DAT) into the RegRipper folder.
3. Open Command Prompt.
4. Navigate to the RegRipper directory:



Step 2: Analyse SYSTEM Hive

To extract shutdown time and USB device information:

```
rip.exe -r SYSTEM -p shutdown
```

```
C:\Users\prath\OneDrive\Desktop\Registry\RegRipper3.0-master\RegRipper3.0-master>rip.exe -r SYSTEM -p shutdown
Launching shutdown v.20200518
shutdown v.20200518
(System) Gets ShutdownTime value from System hive

ControlSet001\Control\Windows key, ShutdownTime value
LastWrite time: 2026-02-15 03:32:46Z
ShutdownTime  : 2026-02-15 03:32:46Z
```

rip.exe -r SYSTEM -p usbstor

```
C:\Users\prath\OneDrive\Desktop\Registry\RegRipper3.0-master\RegRipper3.0-master>rip.exe -r SYSTEM -p usbstor
Launching usbstor v.20200515
usbstor v.20200515
(System) Get USBStor key info

USBStor
ControlSet001\Enum\USBStor

Disk&Ven_hp&Prod_v236w&Rev_PMAP [2026-02-04 07:42:28]
  S/N: 0701093F0C51CB26&0 [2026-02-04 07:42:28Z]
  Device Parameters LastWrite: [2026-02-04 07:42:28Z]
  Properties LastWrite       : [2026-02-04 07:42:28Z]
    FriendlyName          : hp v236w USB Device
    First InstallDate     : 2026-02-04 07:42:28Z
    InstallDate           : 2026-02-04 07:42:28Z
    Last Arrival          : 2026-02-04 07:42:28Z
    Last Removal          : 2026-02-04 09:44:51Z

Disk&Ven__USB&Prod__Sandisk_3.2Gen1&Rev_1.00 [2026-02-02 09:43:00]
  S/N: 0401b6c9cde99db7955fe35ac741cf8c8de7ac3d2dc1b3d612cbd6924309c89 [2026-02-02 09:43:00Z]
  Device Parameters LastWrite: [2026-02-02 09:43:00Z]
  Properties LastWrite       : [2026-02-02 09:43:00Z]
    FriendlyName          :  USB  SanDisk 3.2Gen1 USB Device
    First InstallDate     : 2026-02-02 09:43:00Z
    InstallDate           : 2026-02-02 09:43:00Z
    Last Arrival          : 2026-02-02 09:43:00Z
    Last Removal          : 2026-02-02 10:05:19Z
  S/N: 0401dcefd5e28944f32a932630791f0a0cdd01fc745f44768bacca9372276 44 [2026-02-14 13:37:07Z]
  Device Parameters LastWrite: [2026-01-05 13:59:21Z]
  Properties LastWrite       : [2026-01-05 13:59:22Z]
    FriendlyName          :  USB  Sandisk 3.2Gen1 USB Device
    First InstallDate     : 2026-01-05 13:59:21Z
    InstallDate           : 2026-01-05 13:59:21Z
    Last Arrival          : 2026-02-14 13:37:07Z
    Last Removal          : 2026-02-14 14:38:19Z
```

Artifacts Extracted:

- Last system shutdown time
- USB device history
- Vendor ID and Product ID
- Serial numbers
- First/Last connection timestamps

Step 3: Analyse SOFTWARE Hive

To extract installed applications and OS information:

rip.exe -r SOFTWARE -p uninstall

```
C:\Users\prath\OneDrive\Desktop\Registry\RegRipper3.0-master\RegRipper3.0-master>rip.exe -r SOFTWARE -p uninstall
Launching uninstall v.20200525
uninstall v.20200525
(Software, NTUSER.DAT) Gets contents of Uninstall keys from Software, NTUSER.DAT hives

Uninstall
Microsoft\Windows\CurrentVersion\Uninstall

2026-02-16 06:06:52Z
  Where Winds Meet

2026-02-15 03:42:57Z
  Office 16 Click-to-Run Extensibility Component v.16.0.19628.20204

2026-02-15 03:41:50Z
  Microsoft Office Home and Student 2021 - en-us v.16.0.19628.20204
  Microsoft OneNote - en-us v.16.0.19628.20204

2026-02-15 03:04:34Z
  Red Dead Redemption 2

2026-02-13 02:56:22Z
  Recuva v.1.54

2026-02-11 05:49:20Z
  Microsoft OneDrive v.26.007.0112.0002
```

rip.exe -r SOFTWARE -p winver

```
C:\Users\prath\OneDrive\Desktop\Registry\RegRipper3.0-master\RegRipper3.0-master>rip.exe -r SOFTWARE -p winver
Launching winver v.20200525
winver v.20200525
(Software) Get Windows version & build info

ProductName             Windows 10 Home Single Language
ReleaseID               2009
BuildLab                26100.ge_release.240331-1435
BuildLabEx              26100.1.amd64fre.ge_release.240331-1435
CompositionEditionID    Core
RegisteredOwner         prathapm2016@gmail.com
UBR                     7840
InstallDate             2025-03-10 01:59:41Z
InstallTime             2025-03-10 01:59:41Z
UBR                     7840
```

Artifacts Extracted:

- Installed programs
- Installation dates
- Publisher information
- Operating system version

Step 4: Analyse NTUSER.DAT

To extract user activity artifacts:

rip.exe -r NTUSER.DAT -p userassist

```
C:\Users\prath\OneDrive\Desktop\Registry\RegRipper3.0-master\RegRipper3.0-master>rip.exe -r NTUSER.DAT -p userassist
Launching userassist v.20170204
UserAssist
Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist
LastWrite Time 2025-03-10 02:11:52Z

{9E04CAB2-CC14-11DF-BB8C-A2F1DED72085}

Value names with no time stamps:

{A3D53349-6E61-4557-8FC7-0028EDCEEBF6}

Value names with no time stamps:

{B267E3AD-A825-4A09-82B9-EEC22AA3B847}

{BCB48336-4DDD-48FF-BB0B-D3190DACB3E2}

Value names with no time stamps:

{CAA59E3C-4792-41A5-9909-6A6A8D32490E}

Value names with no time stamps:

{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}
2026-02-17 08:05:56Z
  {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\cmd.exe (6)
2026-02-17 07:45:34Z
  Microsoft.VisualStudioCode (14)
2026-02-17 07:44:58Z
  {6D809377-6AF0-444B-8957-A3773F02200E}\HxD\HxD.exe (8)
2026-02-17 06:46:09Z
  Microsoft.ScreenSketch_8wekyb3d8bbwe!App (26)
2026-02-17 06:23:48Z
  Microsoft.Office.POWERPNT.EXE.15 (2)
2026-02-17 06:03:12Z
  {6D809377-6AF0-444B-8957-A3773F02200E}\Oracle\VirtualBox\VirtualBox.exe (1)
2026-02-17 05:34:47Z
  5319275A.WhatsAppDesktop_cv1g1gvanyjgm!App (12)
  {6D809377-6AF0-444B-8957-A3773F02200E}\WindowsApps\5319275A.WhatsAppDesktop_2.2603.102.0_x64__cv1g1gvanyjgm\WhatsApp.Root.exe (1)
2026-02-17 05:33:36Z
  OperaSoftware.OperaGXWebBrowser.1741573581 (56)
2026-02-16 13:50:40Z
  {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\WindowsPowerShell\v1.0\powershell.exe (3)
2026-02-16 13:18:15Z
  {F38BF404-1D43-42F2-9305-67DE0B28FC23}\regedit.exe (2)
2026-02-16 13:18:08Z
  Microsoft.XboxGamingOverlay_8wekyb3d8bbwe!App (47)
2026-02-16 13:15:16Z
  MSEdge (2)
2026-02-16 13:14:57Z
  D:\Games\Forza Horizon 4\ForzaHorizon4.exe (6)
```

rip.exe -r NTUSER.DAT -p recentdocs

```
Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\Folder
LastWrite Time 2026-02-17 08:17:42Z
MRUListEx = 16,15,18,17,24,28,3,22,11,0,23,29,1,20,25,5,4,10,19,8,14,13,27,26,2,12,9,6,7,21
  16 = Downloads
  15 = bitlocker_gui_sim
  18 = RamAnalysisTool
  17 = Recovered
  24 = RegistryFiles-1
  28 = Understanding
  3 = Lab 4_Regisstry
  22 = Removable Disk (E:)
  11 = core
  0 = gui
  23 = USB (G:)
  29 = 2026-07
  1 = Screenshots
  20 = Document
  25 = Activity 1
  5 = Lab 3
  4 = Lab  3A
  10 = Lab3
  19 = exiftool-13.50_64
  8 = BlackMythWukong
  14 = BITlocker
  13 = PythonScripts
  27 = 2026-06
  26 = docs
  2 = URLyzer
  12 = 2026-05
  9 = Video
  6 = Browser Forensics
  7 = Digital Evidence forms
  21 = volatility-master
```

rip.exe -r NTUSER.DAT -p run

```
C:\Users\prath\OneDrive\Desktop\Registry\RegRipper3.0-master\RegRipper3.0-master>rip.exe -r NTUSER.DAT -p run
Launching run v.20200511
run v.20200511
(Software, NTUSER.DAT) [Autostart] Get autostart key contents from Software hive

Software\Microsoft\Windows\CurrentVersion\Run
LastWrite Time 2026-02-17 06:33:21Z
  Opera GX Browser Assistant - C:\Users\prath\AppData\Local\Programs\Opera GX\assistant\browser_assistant.exe
  Docker Desktop - C:\Program Files\Docker\Docker\Docker Desktop.exe
  MirrorTo - C:\Program Files (x86)\iMyFone\iMyFone MirrorTo\luancher.exe
  OneDrive - "C:\Program Files\Microsoft OneDrive\OneDrive.exe" /background
  utweb - "C:\Users\prath\AppData\Roaming\uTorrent Web\utweb.exe" /MINIMIZED
  EpicGamesLauncher - "C:\Program Files (x86)\Epic Games\Launcher\Portal\Binaries\Win64\EpicGamesLauncher.exe" -silent -launchcontext=boot
  Opera GX Stable - C:\Users\prath\AppData\Local\Programs\Opera GX\opera.exe
  Discord - "C:\Users\prath\AppData\Local\Discord\Update.exe" --processStart Discord.exe
  Teams - "C:\Users\prath\AppData\Local\Microsoft\WindowsApps\MSTeams_8wekyb3d8bbwe\ms-teams.exe" msteams:system-initiated
  RiotClient - C:\Riot Games\Riot Client\RiotClientServices.exe --launch-background-mode
  Steam - "C:\Program Files (x86)\Steam\steam.exe" -silent
  Adobe Acrobat Synchronizer - "C:\Program Files\Adobe\Acrobat DC\Acrobat\AdobeCollabSync.exe"

Software\Microsoft\Windows\CurrentVersion\Run has no subkeys.

Software\Wow6432Node\Microsoft\Windows\CurrentVersion\Run not found.

Software\Microsoft\Windows\CurrentVersion\RunOnce
LastWrite Time 2026-02-01 09:24:13Z
Software\Microsoft\Windows\CurrentVersion\RunOnce has no values.
Software\Microsoft\Windows\CurrentVersion\RunOnce has no subkeys.

Software\Microsoft\Windows\CurrentVersion\RunServices not found.

Software\Microsoft\Windows\CurrentVersion\RunServicesOnce not found.

Software\Microsoft\Windows NT\CurrentVersion\Terminal Server\Install\Software\Microsoft\Windows\CurrentVersion\Run not found.

Software\Microsoft\Windows NT\CurrentVersion\Terminal Server\Install\Software\Microsoft\Windows\CurrentVersion\RunOnce not found.

Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run not found.

Software\Wow6432Node\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run not found.

Software\Microsoft\Windows\CurrentVersion\StartupApproved\Run not found.

Software\Microsoft\Windows\CurrentVersion\StartupApproved\Run32 not found.

Software\Microsoft\Windows\CurrentVersion\StartupApproved\StartupFolder not found.
```

Artifacts Extracted:

- Executed programs
- Execution count
- Recently opened documents
- Startup (Run) entries

Analyzing extracted registry hives using registry explorer tool

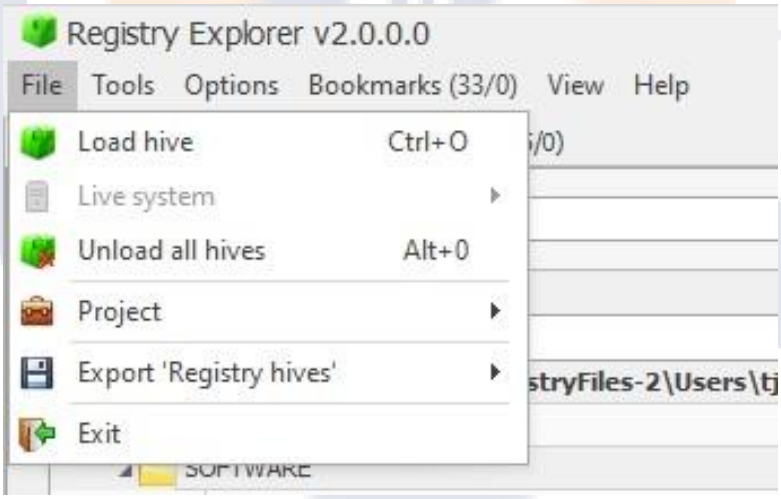
For this project use the registry explorer tool. You can download it from the github link given below.

Registry explorer v2.0.0 (https://ericzimmerman.github.io/#!index.md)

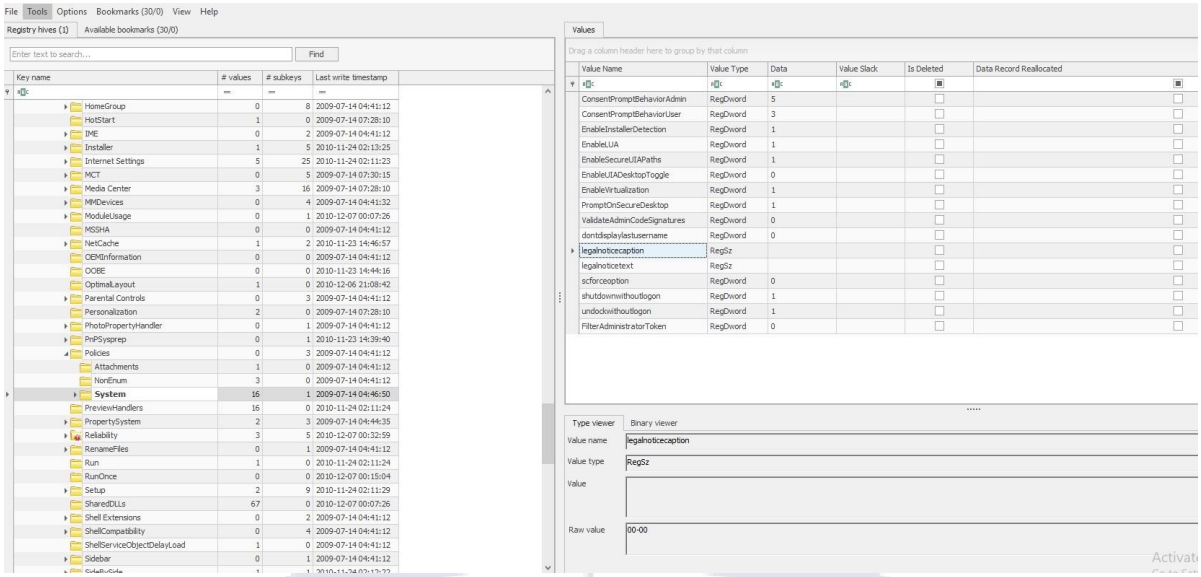
1. Download the file called "RegistryFiles-1.zip" and extract the contents of the compressed file to your desktop.'

2. Right-click the registry explorer tool and select run as administrator.

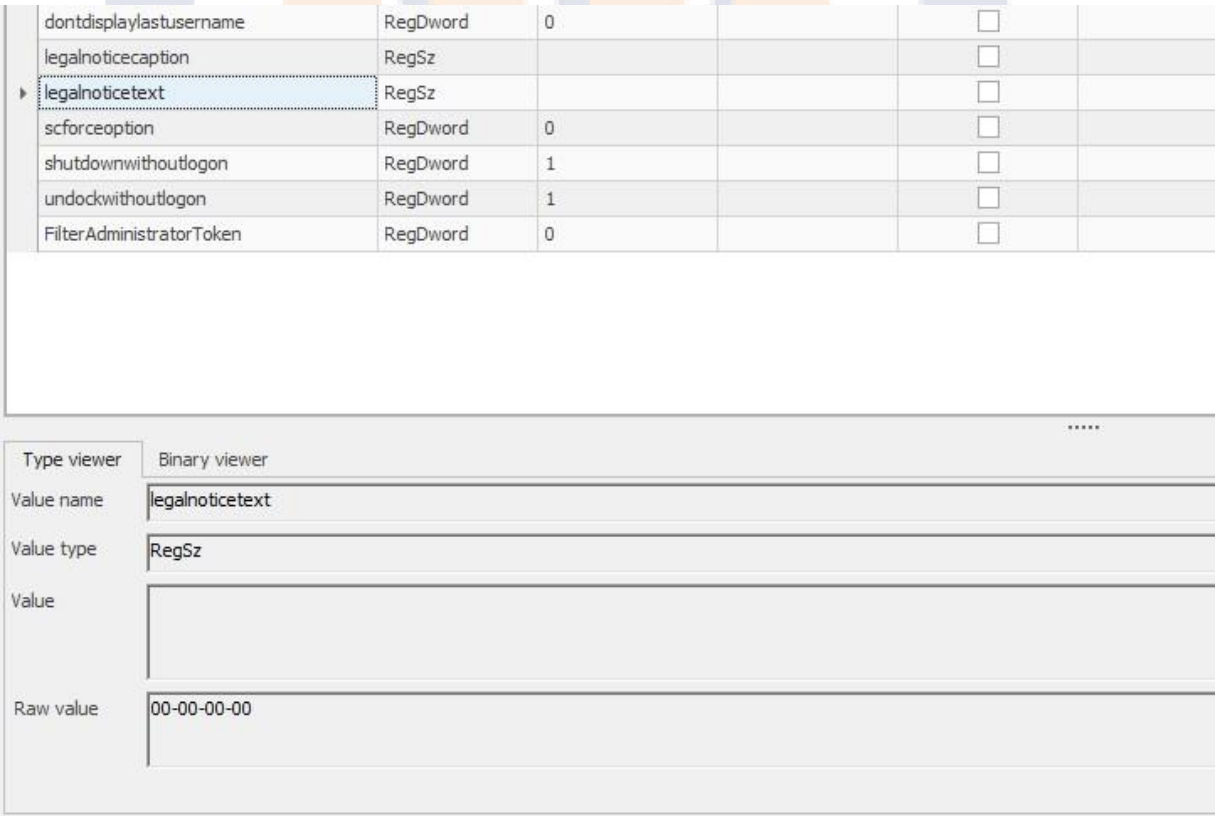
3. To load a hive into registry explorer, select file, Load Hive, and select the path to the "software" hive present in the RegistryFile-1 folder.



4. Confirm the logon banner contained within the Windows Registry of the software hive by navigating down to the following Registry key: \Microsoft\Windows\CurrentVersion\Policies\System



5. After navigating down to the key, the path will be displayed. Notice two keys: legalnoticecaption and legalnoticetext. The former would contain the text value, which appears in the title bar of the consent banner. The latter is the actual message contained within the body of the consent banner.



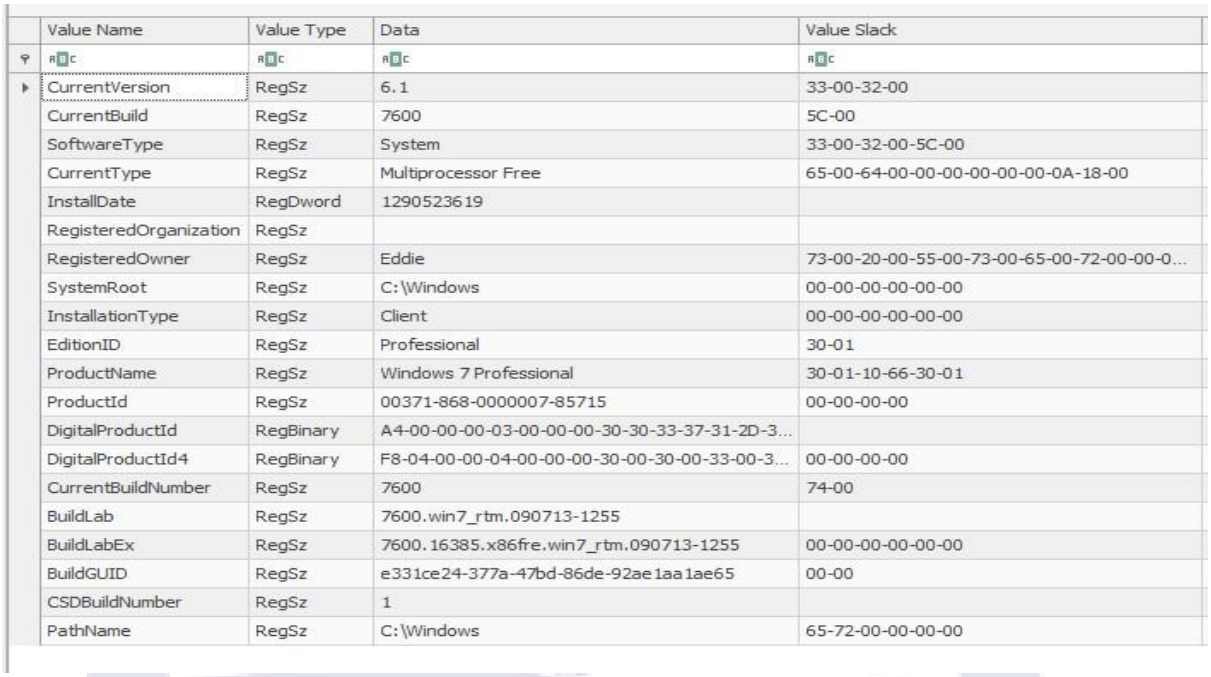
6. What consent banner is shown on this computer?

   Answer:- No consent banner is displayed based on the empty values in this Registry file. (In this hive, the consent banner has been removed and nothing will be displayed at logon. The absence of the banner may cause legal concerns during the examination of corporate assets. In this example the absence of data is a finding.)

7. Navigate to the following key to identify the installation information for the versions of Windows:

Microsoft\Windows NT\CurrentVersion

8. The results of the navigation should appear in a window identical to what is shown



Value Name	Value Type	Data	Value Slack
CurrentVersion	RegSz	6.1	33-00-32-00
CurrentBuild	RegSz	7600	5C-00
SoftwareType	RegSz	System	33-00-32-00-5C-00
CurrentType	RegSz	Multiprocessor Free	65-00-64-00-00-00-00-00-00-0A-18-00
InstallDate	RegDword	1290523619	
RegisteredOrganization	RegSz		
RegisteredOwner	RegSz	Eddie	73-00-20-00-55-00-73-00-65-00-72-00-00-0...
SystemRoot	RegSz	C:\Windows	00-00-00-00-00-00
InstallationType	RegSz	Client	00-00-00-00-00-00
EditionID	RegSz	Professional	30-01
ProductName	RegSz	Windows 7 Professional	30-01-10-66-30-01
ProductId	RegSz	00371-868-0000007-85715	00-00-00-00
DigitalProductId	RegBinary	A4-00-00-00-03-00-00-00-30-30-33-37-31-2D-3...	
DigitalProductId4	RegBinary	F8-04-00-00-04-00-00-00-30-00-30-00-33-00-3...	00-00-00-00
CurrentBuildNumber	RegSz	7600	74-00
BuildLab	RegSz	7600.win7_rtm.090713-1255	
BuildLabEx	RegSz	7600.16385.x86fre.win7_rtm.090713-1255	00-00-00-00-00-00
BuildGUID	RegSz	e331ce24-377a-47bd-86de-92ae1aa1ae65	00-00
CSDBuildNumber	RegSz	1	
PathName	RegSz	C:\Windows	65-72-00-00-00-00

9. What is the name of the Windows product?

   Answer:- The official name is Windows 7 Professional

10. What is the product ID number?

    Answer:- The product ID number is 00371-868-0000007-85715

11. In what directory on the system is the operating system running? **Answer:-** The operating system is using C:\Windows.

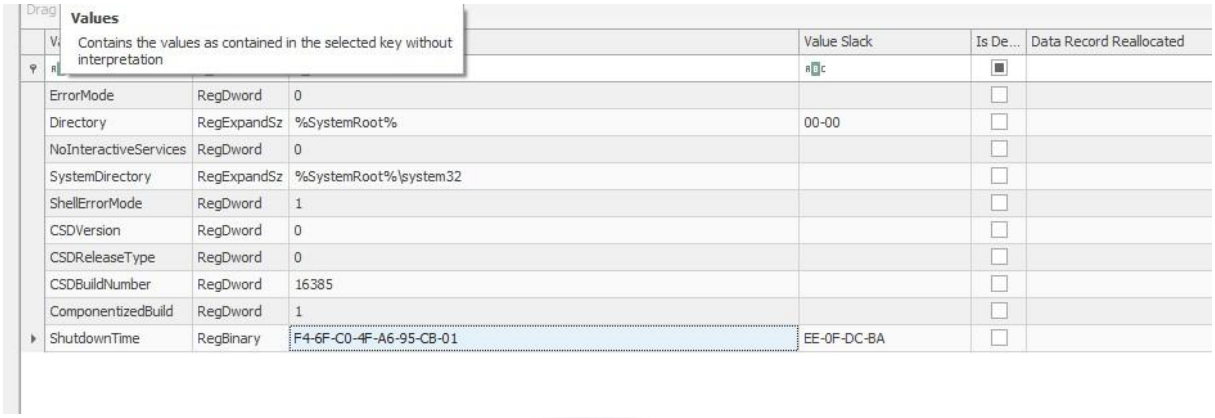
12. When was the operating system installed?

    Answer:- Based on the Registry, the installation date is listed as 1290523619 (, which is a timestamp recorded in Epoch time. Using a resource, such as www.epochconverter.com, converts the value to November 23, 2010 at 14:46:59 GMT.)

13. Load the "system" hive by following the instructions in step 3. What was the last shutdown time of the target system?

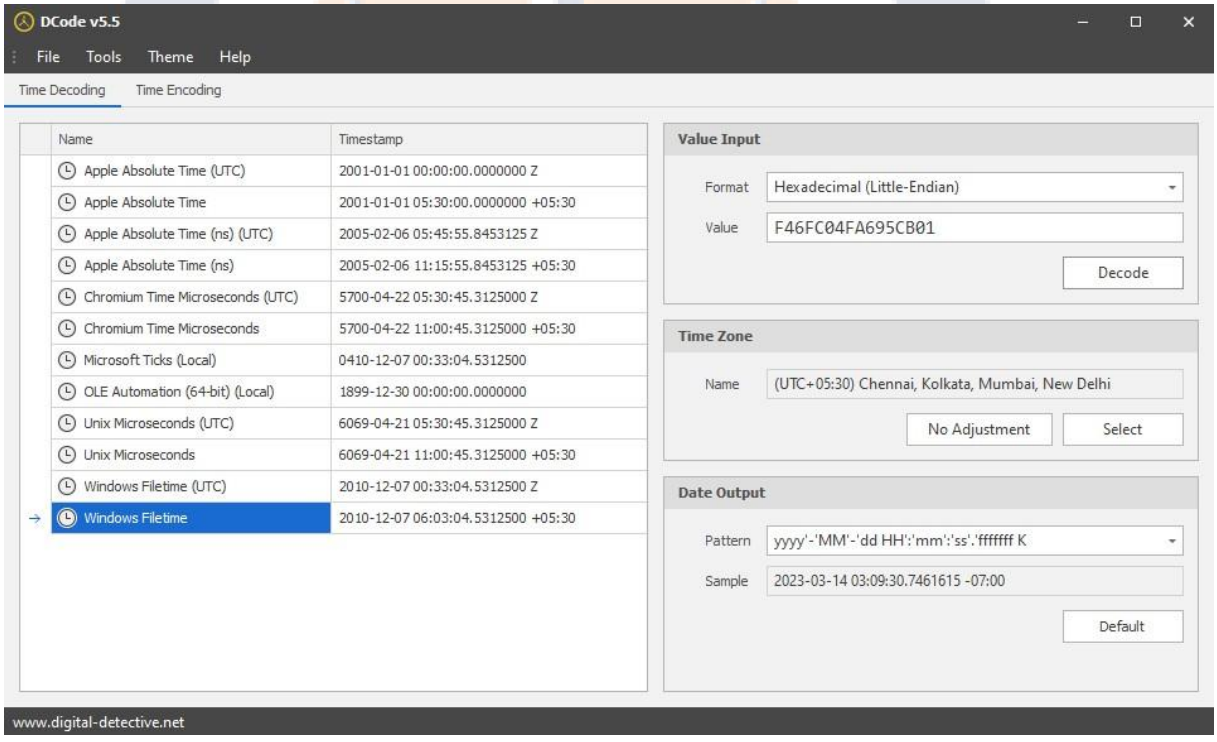
    Answer:- Open the "SYSTEM" hive using registry explorer and find the "CurrentControlSet". Navigate to the following subkey:

CurrentControlSet001\Control\Windows

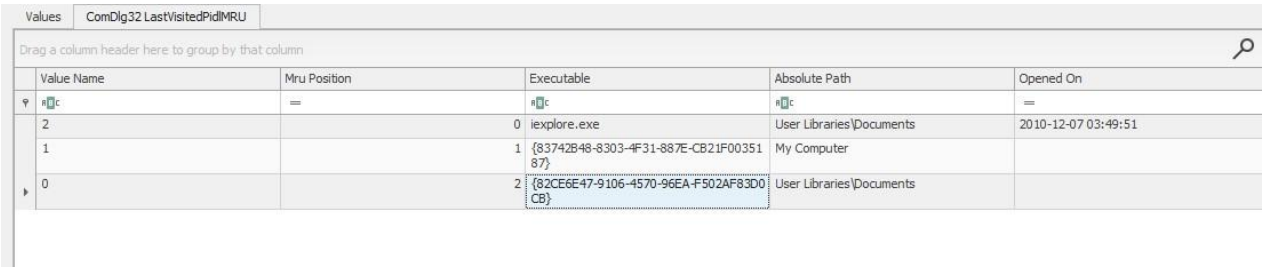


			Value Slack	Is De...	Data Record Reallocated
ErrorMode	RegDword	0			
Directory	RegExpandSz	%SystemRoot%	00-00		
NoInteractiveServices	RegDword	0			
SystemDirectory	RegExpandSz	%SystemRoot%\system32			
ShellErrorMode	RegDword	1			
CSDVersion	RegDword	0			
CSDReleaseType	RegDword	0			
CSDBuildNumber	RegDword	16385			
ComponentizedBuild	RegDword	1			
ShutdownTime	RegBinary	F4-6F-C0-4F-A6-95-CB-01	EE-0F-DC-BA		

14. Using the Dcode tool, convert the hexadecimal time to IST. You can download the tool from the following link - Dcode v5.5 (https://www.digital-detective.net/dcode/).The input format is Hexadecimal (Little Endian), Time Zone as New Delhi and click decode.



15. Identify the executable files that have been executed sometime back in the target system.Open the “NTUSER.DAT” found on the user profile and navigate to the given subkey: Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPIDlMRU



Value Name	Mru Position	Executable	Absolute Path	Opened On
2	0	iexplore.exe	User Libraries\Documents	2010-12-07 03:49:51
1	1	{83742B48-8303-4F31-887E-CB21F00351 87}	My Computer	
0	2	{82CE6E47-9106-4570-96EA-F502AF83D0 CB}	User Libraries\Documents	

16. Identifying the files that have been recently accessed. Open the “NTUSER.DAT” found on the user profile and navigate to this subkey: Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs

RecentDocs	2	Documents	Documents.lnk	0	2010-12-07 03:49:51	2010-12-07 03:49:51
RecentDocs	6	Jailbreak iPhone 4 - How to Jailbreak iPhone 4.mht	Jailbreak iPhone 4 - How to Jailbreak iPhone 4.lnk	1		2010-12-07 03:49:51
RecentDocs	5	Iphone 4 Jailbreak.mht	Iphone 4 Jailbreak.lnk	2		
RecentDocs	4	Computer	Computer.lnk	3		
RecentDocs	3	plumbing problem.docx	plumbing problem.lnk	4		2010-12-06 22:42:37
RecentDocs	1	iPhones.xlsx	iPhones.lnk	5		2010-12-06 21:16:37
RecentDocs	0	System and Security	System and Security.lnk	6		
Folder	1	Documents	Documents.lnk	0	2010-12-07 03:49:51	
Folder	2	Computer	Computer.lnk	1		
Folder	0	System and Security	System and Security.lnk	2		
.xlsx	0	iPhones.xlsx	iPhones.lnk	0	2010-12-06 21:16:37	
.mht	1	Jailbreak iPhone 4 - How to Jailbreak iPhone 4.mht	Jailbreak iPhone 4 - How to Jailbreak iPhone 4.lnk	0	2010-12-07 03:49:51	
.mht	0	Iphone 4 Jailbreak.mht	Iphone 4 Jailbreak.lnk	1		
.docx	0	plumbing problem.docx	plumbing problem.lnk	0	2010-12-06 22:42:37	

17. Once you are done with your analysis, right click the hive that you want to remove from the registry hives tab and select remove hive.